



ST BENEDICT'S

SUBJECT	PHYSICS FINAL EXAM
GRADE	11
EXAMINER	Mr Hilder
MODERATOR	Mr McCoy
DATE	21 November 2022
MARKS	200
DURATION	180 minutes

MARK
TOTAL

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LEARNER NAME:

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 30 pages and a Data Sheet of 2 pages, please detach the Data Sheet. Please check that your question paper is complete.
2. Read the questions carefully.
3. Answer **ALL** the questions on the question paper and hand it in at the end of the assessment. Remember to write your name and surname in the space provided above.
4. Use the data and formulae sheet whenever necessary.
5. Show your working in all calculations.
6. Units need not be included in the working of calculations, but appropriate units should be shown in the answer.
7. Answers must be expressed in decimal format, not left as proper fractions.
8. Where appropriate, express answers to TWO decimal places.
9. It is in your own interest to write legibly and to present your work neatly.
10. Blank pages are included at the end of the paper (page 29 & 30). If you run out of space for a question, use this page. Clearly indicate the question number of your answer should you use this extra space.

FOR OFFICE USE ONLY: MARKER TO ENTER MARKS

Question	1	2	3	4	5	6	7	8	9	10	11	12	Total
Mark													
Moderated Mark													
Question Total	20	18	17	14	16	11	20	11	17	10	28	18	200

Comment:

QUESTION 1 MULTIPLE CHOICE**20 MARKS**

Answer these questions on the multiple-choice answer sheet below. Make a cross (X) in the box corresponding to the letter that you consider to be correct.

A	B	☒	D
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Here the option C has been marked as an example

1.1	A	B	C	D
1.2	A	B	C	D
1.3	A	B	C	D
1.4	A	B	C	D
1.5	A	B	C	D
1.6	A	B	C	D
1.7	A	B	C	D
1.8	A	B	C	D
1.9	A	B	C	D
1.10	A	B	C	D

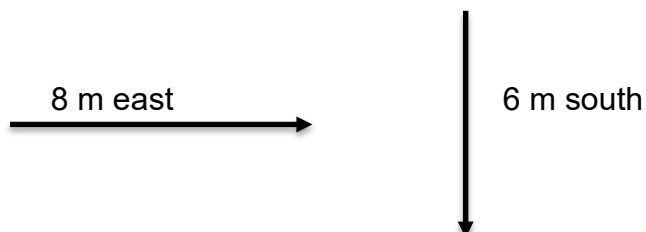
1.1 Kinetic energy can be expressed mathematically as $E_k = \frac{1}{2}mv^2$.

This equation consists of:

(2)

- A one vector quantity and two scalar quantities
- B two vector quantities and one scalar quantity
- C three vector quantities
- D three scalar quantities

1.2 The diagram below shows two vectors.



What is the magnitude of the resultant of these two vectors?

(2)

- A 10 m
- B 12 m
- C 14 km
- D 16 km

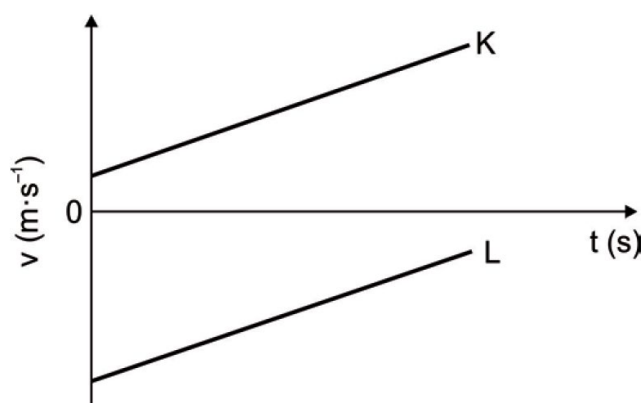
- 1.3 A car is moving with a velocity $\mathbf{v}$. The driver slams on brakes and the car stops after traveling a certain distance $\mathbf{d}$.
If the driver was traveling at triple the initial velocity, $3\mathbf{v}$ and braked with the same force, how would the car's stopping distance compare to the first distance? (2)

- A $\frac{1}{3}d$
B $3d$
C $6d$
D $9d$

- 1.4 A person stands on a bathroom scale in a stationary elevator. The reading on the scale is **490 N**. When the elevator is in motion, the reading on the scale changes to **470 N**. Which one of the following combinations best describe the direction of the motion and the direction of the acceleration of the elevator during the motion? (2)

	Direction of motion	Direction of acceleration
A	Upwards	Upwards
B	Downwards	Downwards
C	Upwards	Downwards and then upwards
D	Downwards	Upwards and then downwards

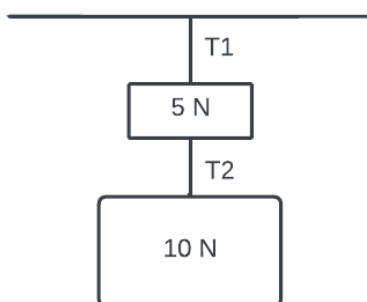
- 1.5 The velocity vs time graph shows the motion of two different objects sliding in a straight line across a horizontal surface.



For which object could friction be the cause of the change in velocity? (2)

- A K
B L
C L and K
D Neither

- 1.6 Two blocks of weight 5 N and 10 N are attached to two ropes, T1 and T2 as shown below.



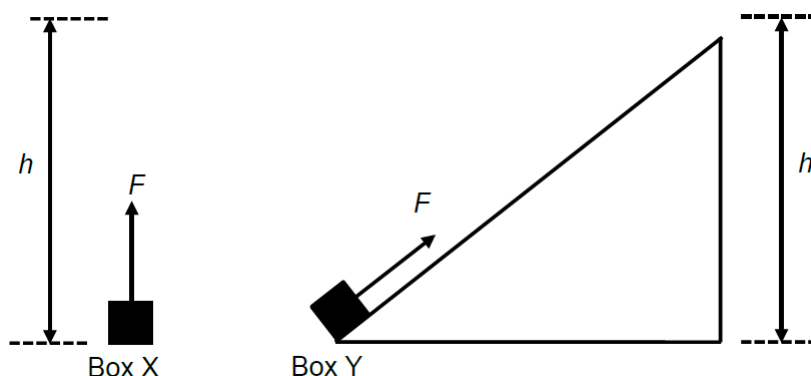
The blocks hang vertically at rest. The masses of the ropes are negligible. What is the tension in rope T1 and in rope T2?

(2)

	Tension in T1	Tension in T2
A	15 N	15 N
B	10 N	15 N
C	5 N	15 N
D	15 N	10 N

- 1.7 Two boxes X and Y have the same mass. Box X is lifted through a vertical height h by a force of magnitude F . Box Y is pulled up a slope by a force of the same magnitude to reach the same vertical height, as shown.

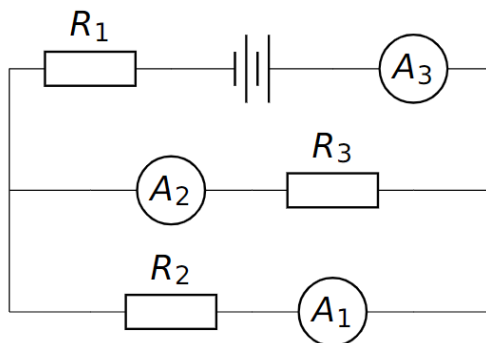
(2)



	Change in gravitational potential energy of the boxes.	Amount of work done by F .
A	Is the same for box X and Y	Is the same on X and Y
B	Is the same of box X and Y	Is more for box Y than for box X.
C	Is less for box Y than for box X	Is less for box Y than for box X
D	Is more for box Y than for box X	Is more for box Y than for box X

1.8 Which equation correctly describes the current in the circuit diagram?

(2)

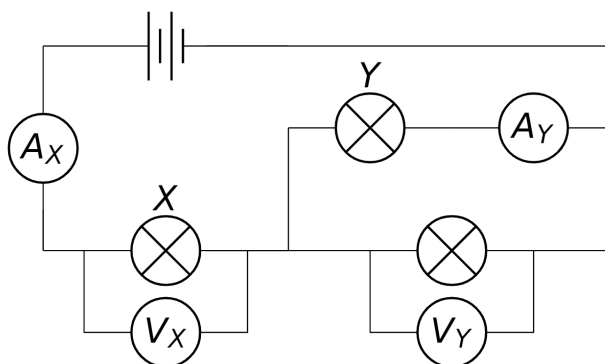


- A $A_1 = A_2 + A_3$
 B $A_2 = A_3 - A_1$
 C $A_3 = A_2 - A_1$
 D $A_2 = A_1 + A_3$

1.9 In the circuit shown below the bulbs are all identical.

Which one of the following correctly gives the ratio of the power dissipated in bulb X compared to the power dissipated in bulb Y.

(2)



	P(X) : P(Y)
A	1 : 4
B	1 : 2
C	4 : 1
D	2 : 1

1.10 Which one of the following types of electromagnetic radiation has the highest energy?

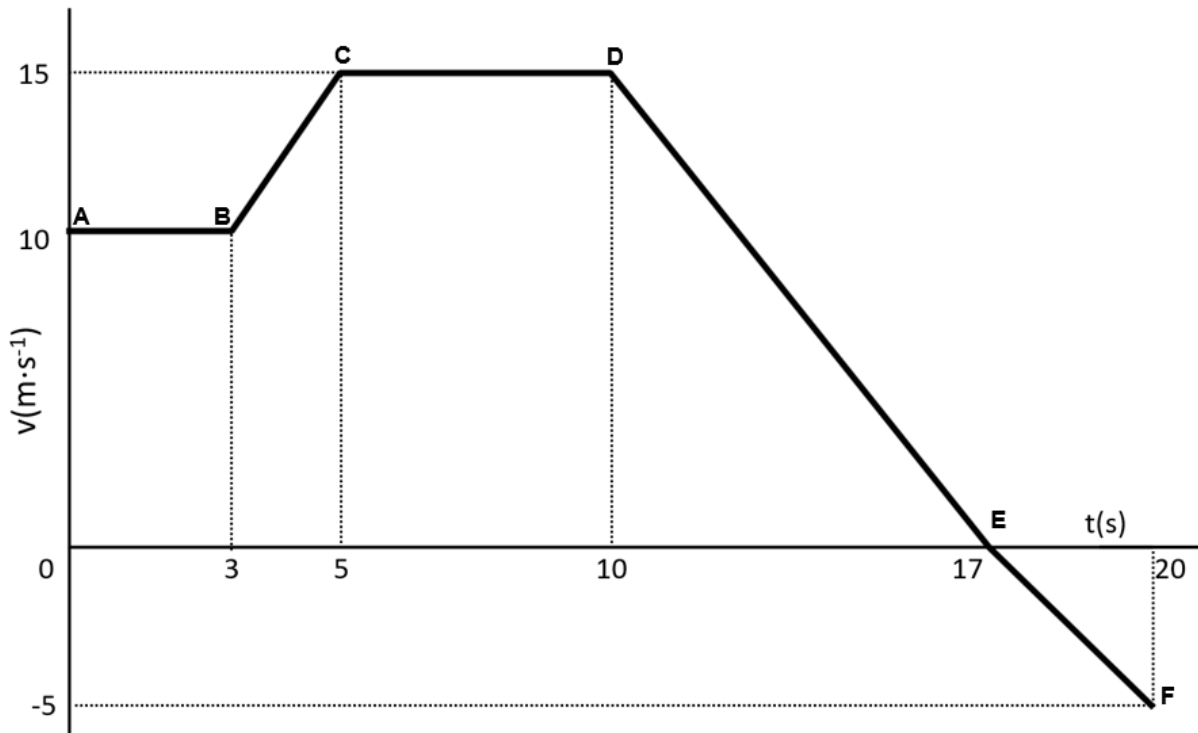
(2)

- A X-rays
 B Gamma rays
 C Infra-red
 D Ultra-violet

QUESTION 2 KINEMATICS GRAPH**18 MARKS**

The motion of an object travelling south in a straight line is represented in the velocity time graph shown below.

Using the graph and without the use of the equations of motion, answer the following questions.



- 2.1 At which point(s) is the object stationary? (2)

- 2.2 At which stage(s) is the object travelling with a constant velocity. (2)

- 2.3 What is the displacement of the object between B and C? (4)

- 2.4 Calculate magnitude of acceleration between D and E. (3)

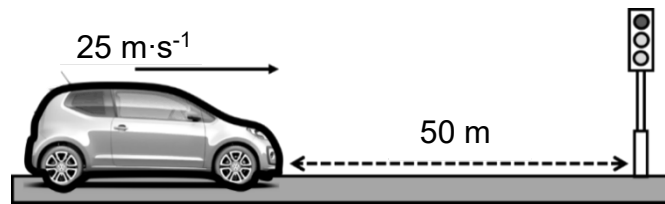
2.5 Describe the motion of the object between D and F. (3)

2.6 Draw a position vs time graph of the motion between B and E on the axis provided. (4)



QUESTION 3 HORIZONTAL MOTION**17 MARKS**

A VW is travelling at a speed of $25 \text{ m}\cdot\text{s}^{-1}$ when a traffic light, 50 m ahead, changes to red. The driver takes 0,35 s to react before slamming on the brakes.



Once the brakes have been applied, the car comes to a stops 3,2 s.

3.1 Define *speed*. (2)

3.2 What was the initial speed of the car in kilometres per hour? (2)

3.3 Calculate the distance the car travels during the reaction time. (2)

3.4 Will the car stop before the traffic light? Show ALL calculations. (4)

- 3.5 The driver's phone had been placed on the passenger seat next to them. When the brakes were applied the phone slid forward landing on the floor in the foot well. Refer to one of Newton's Laws of motion to explain why this happened. (2)

- 3.6 While the WV is waiting for the traffic light to turn green, a BMW approaches from behind with a uniform velocity of $12 \text{ m}\cdot\text{s}^{-1}$. The instant the BMW is directly next to the VW, the traffic light turns green and the VW begins accelerating uniformly for 4 s reaching a velocity of $18 \text{ m}\cdot\text{s}^{-1}$. When the VW reaches $18 \text{ m}\cdot\text{s}^{-1}$ it stops accelerating and maintains this velocity. How far away from the initial traffic light will both cars be when the VW catches up with the BMW? (5)

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QUESTION 4 PROJECTILE MOTION

14 MARKS

A drone is being flown directly upward at a constant velocity of $3 \text{ m}\cdot\text{s}^{-1}$.

As the drone is gaining altitude the camera attached to the bottom of the drone falls off.



It takes 3,6 s to hit the ground below. Air resistance can be ignored

- 4.1 Define *displacement*. (2)

- 4.2 Calculate the time that it will take the camera to reach its maximum height. (3)

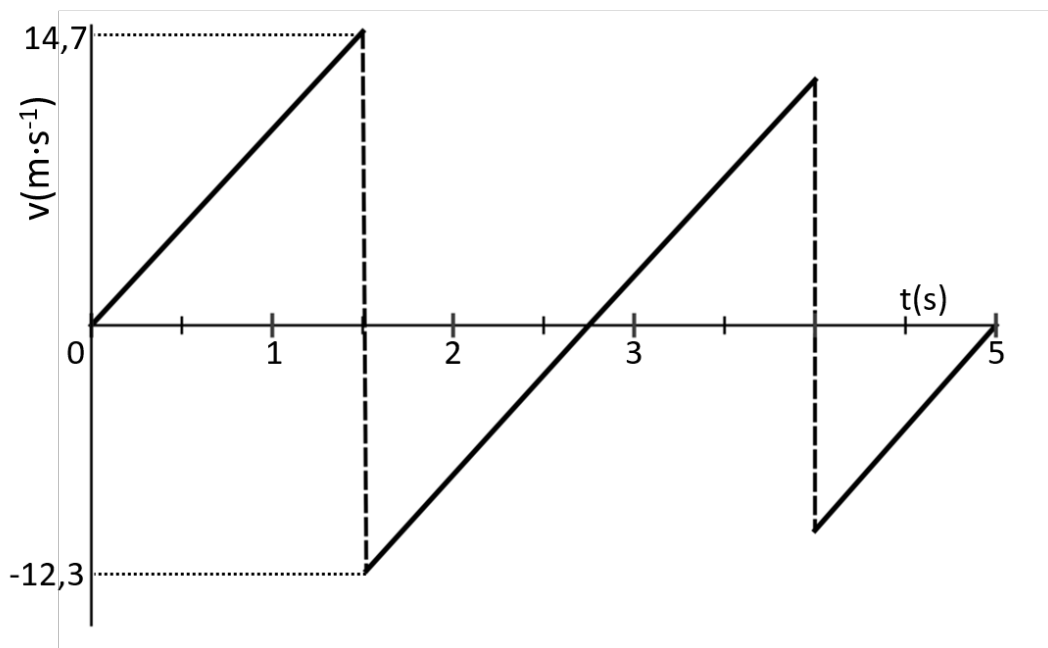
- 4.3 At what height above the ground was the drone when the camera fell off? (3)

- 4.4 Calculate the maximum height reached by the camera above the ground. (4)

- 4.5 In reality air resistance **cannot be ignored**, describe the effect air resistance would have on the distance the camera travels as it falls. (2)

QUESTION 5 BOUNCE**16 MARKS**

A student drops a tennis ball from a certain height. The velocity-time graph represents the motion of the ball as it bounces vertically on a solid floor. The time of bounce is negligible and air resistance may be ignored.



5.1 Define *acceleration*. (2)

5.2 Describe the motion of the ball from 1,5 to 2,75 seconds. (2)

5.3 Calculate the height from which the ball was dropped. (3)

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5.4 Was energy lost during the bounce? Circle YES or NO.

Briefly explain your answer.

(3)

5.5 State the magnitude of the velocity of the ball at 4 s.

(2)

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5.6 Suggest a possible reason for the motion experienced by the ball at 5 s.

(1)

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5.7 Sketch an acceleration versus time graph of the ball from 0s to 4 s.

Indicate the acceleration of the ball when it hits the ground.

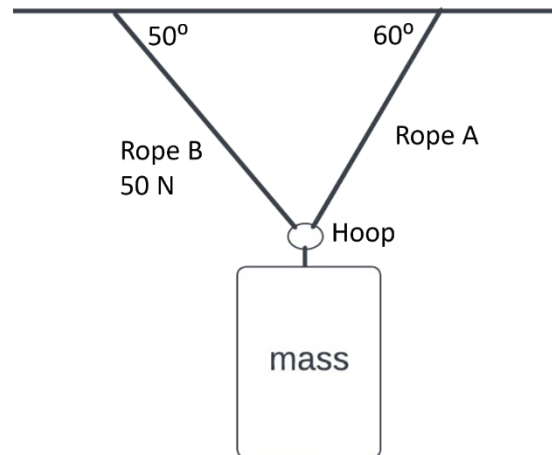
(3)



QUESTION 6 ROPES

11 MARKS

A block is hanging at rest from a hoop, which is attached to the ceiling by two ropes. The ropes make angles to the horizontal ceiling of 50° and 60° as shown in the diagram. The tension in the rope B is 50 N.



- 6.1 Define a *vector*. (2)

- 6.2 Determine the magnitude of the horizontal component of the tension in rope B. (2)

- 6.3 Calculate the magnitude of the tension in the rope A. (3)

6.4 Determine the block's mass.

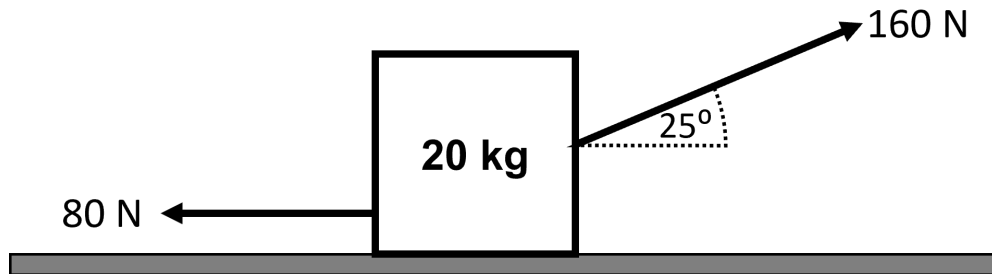
(4)

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QUESTION 7 SLIDING BLOCKS

20 MARKS

- 7.1 A 20 kg metal box is pulled across a rough horizontal surface with a force of 160 N acting at an angle of 25° to the surface. The kinetic frictional force acting on the metal box is 80 N.



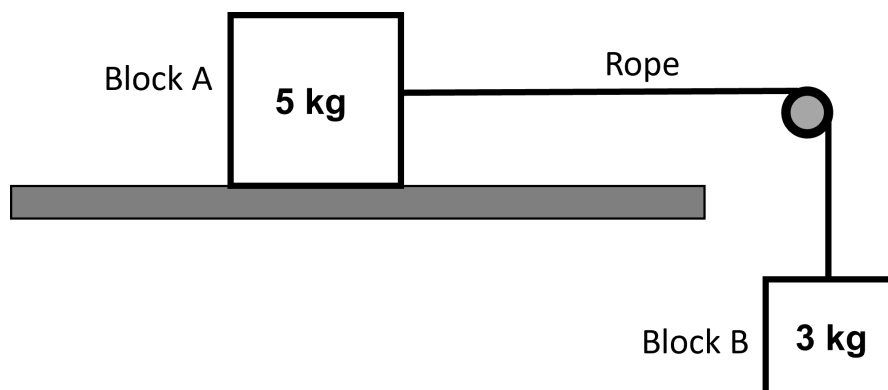
- 7.1.1 Draw a free-body diagram showing all the forces acting on the metal box. (4)

- 7.1.2 State *Newton's second law of motion*. (2)

- 7.1.3 Calculate the magnitude of the acceleration of the metal box. (4)

- 7.1.4 Calculate the magnitude of the normal force that the ground is exerting on the box. (3)

- 7.2 Two blocks are connected by a massless rope. Block A is being pulled across a rough surface by the weight of Block B. The blocks are accelerating clockwise at $2,5 \text{ m}\cdot\text{s}^{-2}$.



- 7.2.1 What is the magnitude of friction acting on Block A? (4)

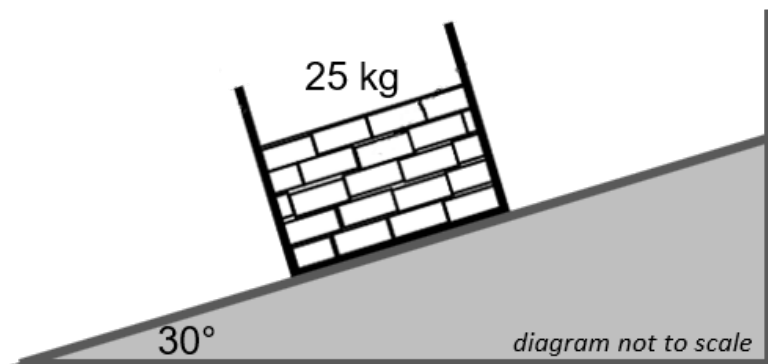
7.2.2 Determine the tension in the Rope.

(3)

QUESTION 8 SLOPE

11 MARKS

8.1 A crate of bricks with a mass of 25 kg is on the verge of slipping when placed on a wooden plane inclined at 30° from the horizontal.



8.1.1 State *Newton's first law*.

(2)

8.1.2 What is the magnitude of the net force ($\mathbf{F}_{\text{net}}$) acting on the container of bricks while it is at rest on the slope?

(1)

8.1.3 Calculate the normal force acting on the box. (2)

8.1.4 Use you answer from QUESTION 8.1.3 to determine the co-efficient of static friction between the crate and the surface of the slope. (3)

8.1.5 What effect would lowering the incline of the slope have on the co-efficient of static friction? INCREASE , DECREASE or REMAINS THE SAME. Circle the correct option. (1)

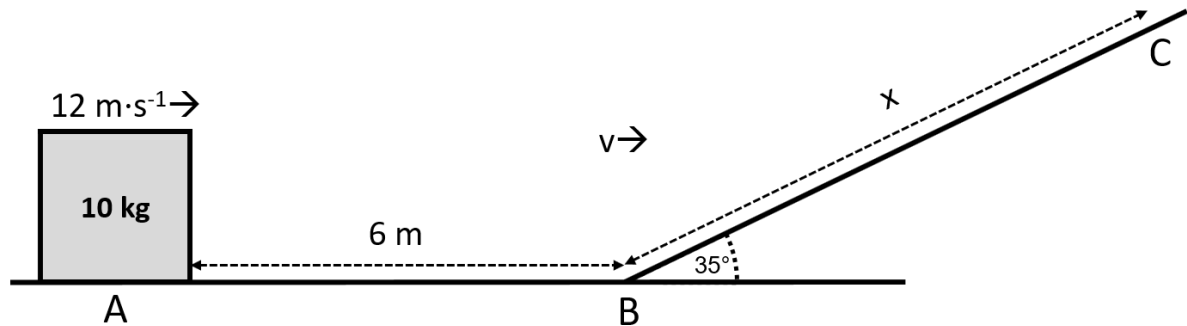
8.1.6 Explain your answer to QUESTION 8.1.5 (2)

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QUESTION 9 WORK-ENERGY

17 MARKS

A block of mass 10 kg is sliding along surface. The horizontal surface from A to B is rough and the surface from B to C is a smooth incline at 35° from the horizontal.



The block is travelling at a speed of $12 \text{ m}\cdot\text{s}^{-1}$ as it passes A.

9.1 Define the term *kinetic energy*. (2)

9.2 Calculate the kinetic energy of the block as it passes A. (3)

The frictional force acting on the block as it slides from A to B is 54,9 N.

9.3 State the *work-energy theorem*. (2)

9.4 Calculate the speed of the block (v) as it reaches B. (4)

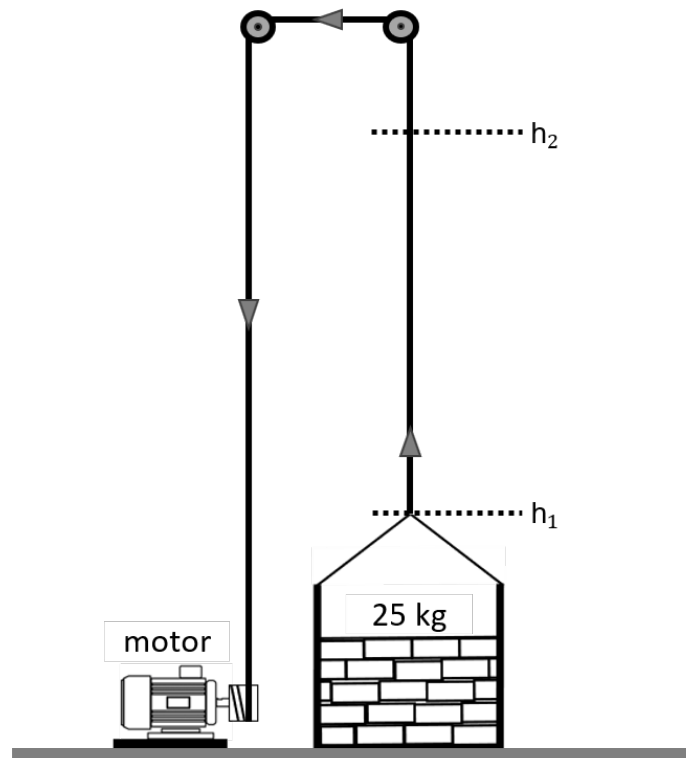
The block slides up the incline from B and comes to rest at C. The frictional force acting on the block as it slides from B to C is negligible.

9.5 Define *mechanical energy* (2)

9.6 Calculate the distance, x , that the block slides up the slope before coming to rest at C. (4)

QUESTION 10 POWER**10 MARKS**

A 1600 Watt electric motor is used to lift a 25 kg box of bricks from h_1 (0 m) to h_2 (18 m) at a constant velocity.



10.1 Calculate the power required to lift the bricks at a constant velocity of $6 \text{ m}\cdot\text{s}^{-1}$. (3)

10.2 Determine the work done by the motor to lift the box 18 m at the stated velocity. (3)

10.3 How efficient is the motor while lifting the box of bricks at $6 \text{ m}\cdot\text{s}^{-1}$?

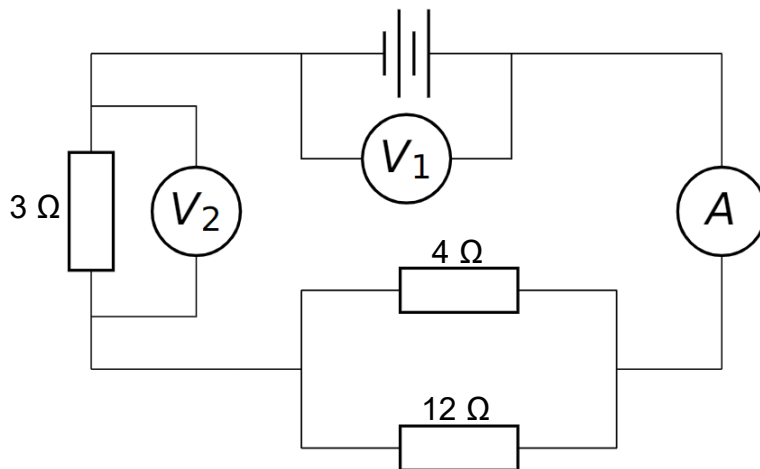
(4)

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QUESTION 11 CIRCUITS

28 MARKS

11.1 A circuit that consists of a battery with an emf of 18 V, three resistors, an ammeter and two voltmeters are set up as shown in the diagram below.



11.1.1 Define *resistance*. (2)

11.1.2 Calculate the effective resistance of the circuit. (4)

The ammeter measures a current of 3 A with the circuit connected as shown.

11.1.3 State *Ohm's Law*. (2)

11.1.4 Determine the current through the $4\ \Omega$ resistor.

(4)

11.1.5 Determine the power dissipated by the $3\ \Omega$ resistor.

(3)

Sometimes resistors burn out while in operation

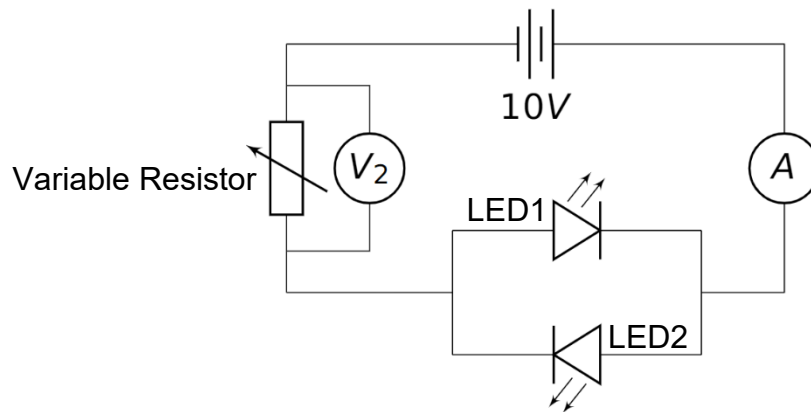
11.1.6 Describe what would happen to the reading on the voltmeter, V_2 , if the $12\ \Omega$ resistor burnt out. Use a suitable equation in your answer.

(3)

11.1.7 Considering the $12\ \Omega$ resistor has burnt out, how much energy would the battery supply to the circuit in 2 minutes?

(4)

- 11.2 A new circuit is built using two LEDs and a variable resistor as shown in the diagram. Both LEDs have a knee voltage of 2,5 V at a current of 20 mA.



- 11.2.1 Which LED will not light up? LED1 or LED2.
Circle the correct option.

(1)

- 11.2.2 Explain your answer to QUESTION 11.2.1.

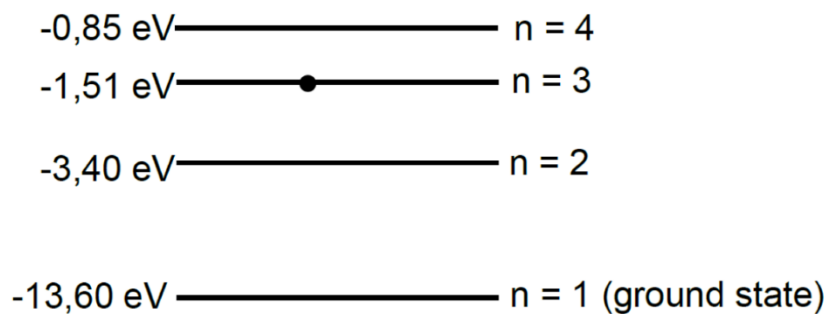
(1)

- 11.2.3 What should the resistance of the variable resistor be set to, in order to provide enough voltage for the LED to operate. (4)

(4)

QUESTION 12 WAVES AND LIGHT**18 MARKS**

- 12.1 The energy levels for a hydrogen atom are represented on the diagram below. The energy levels for a hydrogen atom are represented on the diagram below.



- 12.1.1 An excited electron is in the 3rd energy level as indicated.

How many unique frequencies will be in the emission spectrum as the excited electron returns to the ground state? (1)

- 12.1.2 Calculate the frequency of the photon emitted when the electron transition is from the $n=2$ level to the ground state. (4)

Hydrogen gas in a discharge tube emits light of wavelength 655 nm.

- 12.1.1 Calculate the energy (in eV) that corresponds to this wavelength of light. (4)

PLEASE TURN OVER

12.1.2 Which transition that will result in a wavelength of 655 nm?

(2)

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12.1.3 Why are the energy levels stated as negative values?

(2)

12.2 A FM radio station broadcasts using waves with a wavelength of 3,17 m.

12.2.1 What is the frequency of their transmission?

(3)

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702 Talk Radio broadcasts using AM radio waves which have a frequency in the kHz.

12.2.2 Refer to the properties of FM and AM radio waves to explain why AM is not suitable for music-based radio but can be used for talk radio.

(2)

END OF ASSESMENT

ADDITIONAL SPACE

[illegible]

